

Noise-tailored two-qubit gates on the experimentally anchored noise model: numbers for the three asks

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June 11, 2026

Everything below is computed end-to-end on the experimentally anchored noise model (§1) with the corrected pipeline: blind frequency-comb QNS reconstruction $\rightarrow$ spectra-driven pulse optimization, with the optimizer never seeing ground truth, and every quoted “true” infidelity evaluated exactly on the analytic model. All numbers regenerated 2026-06-11.

1 How much dephasing? The anchored model

The synthetic spectra used previously were replaced by a model anchored to measured silicon and GaAs dephasing data: the six-spectrum structure and charge-noise correlations of the Tarucha-group Si/SiGe pair [Yoneda *et al.*, Nat. Phys. **19**, 1793 (2023)], purified- ^{28}Si coherence as the amplitude anchor, high-frequency exponents from natural-Si spectroscopy [Rojas-Arias *et al.*], nuclear-Larmor-difference lines from GaAs [Malinowski *et al.*, Nat. Nano. **12**, 16 (2017)], and exchange noise at the operating point [Dial *et al.*, PRL **110**, 146804 (2013)]. The “featured” regime adds the line triplet to the monotonic (“bland”) background.

quantity	value	anchor
time unit (min. pulse separation)	$\tau = 25 \text{ ns}$	legacy anchor
T_2^* (both qubits)	$800 \tau = 20 \mu\text{s}$	purified ^{28}Si
Gaussian FID $T_2 = 2/S_{a,a}(0)$	$771 \tau / 770 \tau \approx 19.3 \mu\text{s}$	model output
electrical fraction at $\omega\tau = 0.35$	0.88 (Q1) / 0.80 (Q2)	Yoneda 2023 structure
nuclear-difference lines	$\omega\tau \in \{0.261, 0.273, 0.534\}$ $= 1.66/1.74/3.40 \text{ MHz}, \sigma_{\text{line}}\tau = 0.02$	Malinowski fingerprint (two near-degenerate + one at $2\times$)
ZZ-channel coupling	$S_{12,12}/\sqrt{S_{1,1}S_{2,2}} \approx 0.1, \gamma_J = 0.7$	Dial operating-point J -noise
gate windows studied	$T_G = 40\text{--}2560 \tau = 1\text{--}64 \mu\text{s} = 0.05\text{--}3.3 T_2$	

Table 1: Ask 1 in one table: the dephasing scale and where the gates sit relative to it.

2 Characterization: the reconstruction lands on truth

No-SPAM pipeline (4k shots, t -grain 1600). The full blind reconstruction — harmonic comb inversion, line/tail/head-aware unfolding, DC points from noise-aware slope fits and the double-echo $S_{12,12}(0)$ experiment — is statistically consistent with truth at honestly quoted uncertainties (Table 2). At 64k shots (the SPAM-arm runs) the same pipeline reaches 1–6% median deviations with the same pull structure.

SPAM arms (64k shots). Figure 2 (the four-arm comparison) summarizes the SPAM story at the spectroscopy level: the unmitigated (raw) arm’s SPAM bias is resolvable at **14–34 σ** across channels; the SPAM-mitigated arm is statistically indistinguishable from the SPAM-free reference everywhere; the SPAM-robust protocol is immune by construction but cannot reconstruct $S_{1,12}, S_{2,12}$.

3 Ask 2: do I need all six spectra?

The answer splits cleanly into *error budgeting* and *control design*:

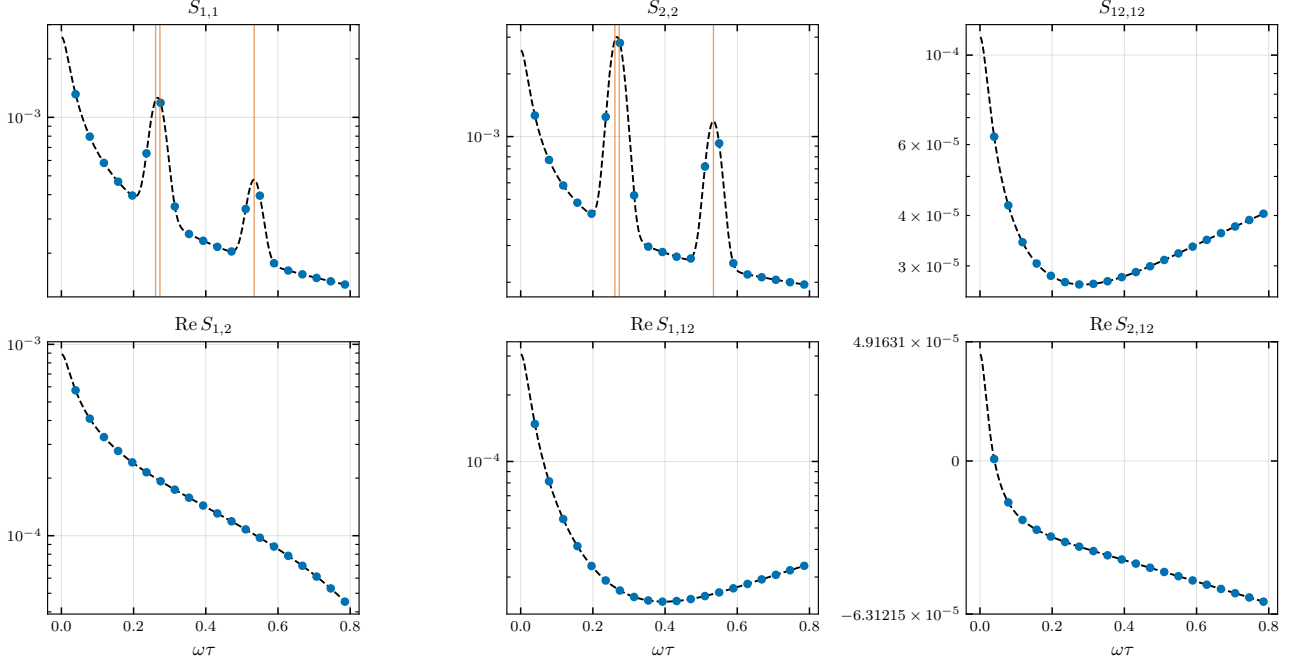


Figure 1: The six anchored spectra (featured regime) and the 20-tooth comb the QNS protocol samples ($T = 160\tau$, teeth at $\omega_k\tau = 2\pi k/160$, DC point from dedicated slope-fit/double-echo experiments). Vertical lines: the nuclear-difference triplet on the single-qubit channels (J -noise is electrical, so $S_{12,12}$ carries no lines).

spectrum	median rel. dev.	median pull	max pull	within 2σ
$S_{1,1}$	4.8%	0.48	2.7	95%
$S_{2,2}$	3.4%	0.85	3.2	95%
$S_{12,12}$	15.3%	0.76	1.7	100%
$S_{1,2}$	6.7%	0.70	1.4	100%
$S_{1,12}$	13.1%	0.88	1.5	100%
$S_{2,12}$	20.8%	0.75	2.1	90%

Table 2: Reconstruction vs. truth for the 4k-shot no-SPAM run (pull = deviation / quoted total uncertainty). Deviations are statistics-dominated and scale as expected from the 64k-shot arms ($\times\sqrt{16}$); the bars are honest (coverage $\geq 90\%$). $S_{12,12}(0)$ is *determined* by the double-echo experiment; the cross-spectra DC points are honestly flagged not-determined at 4k shots.

(a) **For the error budget — yes.** Undecoupled, the two-qubit channels ($S_{12,12}$ and the cross-spectra) carry only 1.2% of the gate error at $T_G = 320\tau$. Under the best decoupling, CDD(7,7), they carry **20.4%**: decoupling suppresses the single-qubit channels so effectively that the two-qubit channels are promoted to a fifth of the residual error budget. Without two-qubit QNS you cannot predict or budget a decoupled gate’s error.

(b) **For the optimizer — largely no, and that is a feature.** We re-ran the blind optimization with progressively reduced spectral knowledge (the ideal benchmark always keeps full truth); Fig. 3(left) and Table 3. The blind NT winner is essentially unchanged down to single-qubit-only knowledge at both gate times: dropping the cross-spectra changes nothing at all, and even dropping $S_{12,12}$ costs only 6% at $T_G = 80\tau$ (and nothing at 320τ) — the single-qubit spectra dominate the design, and the CZ phase constraint pins the y_{12} filter’s DC weight regardless of what the optimizer knows about $S_{12,12}$. In particular, the **SPAM-robust protocol’s four-spectrum set costs nothing at the gate level** — which couples directly into ask 3.

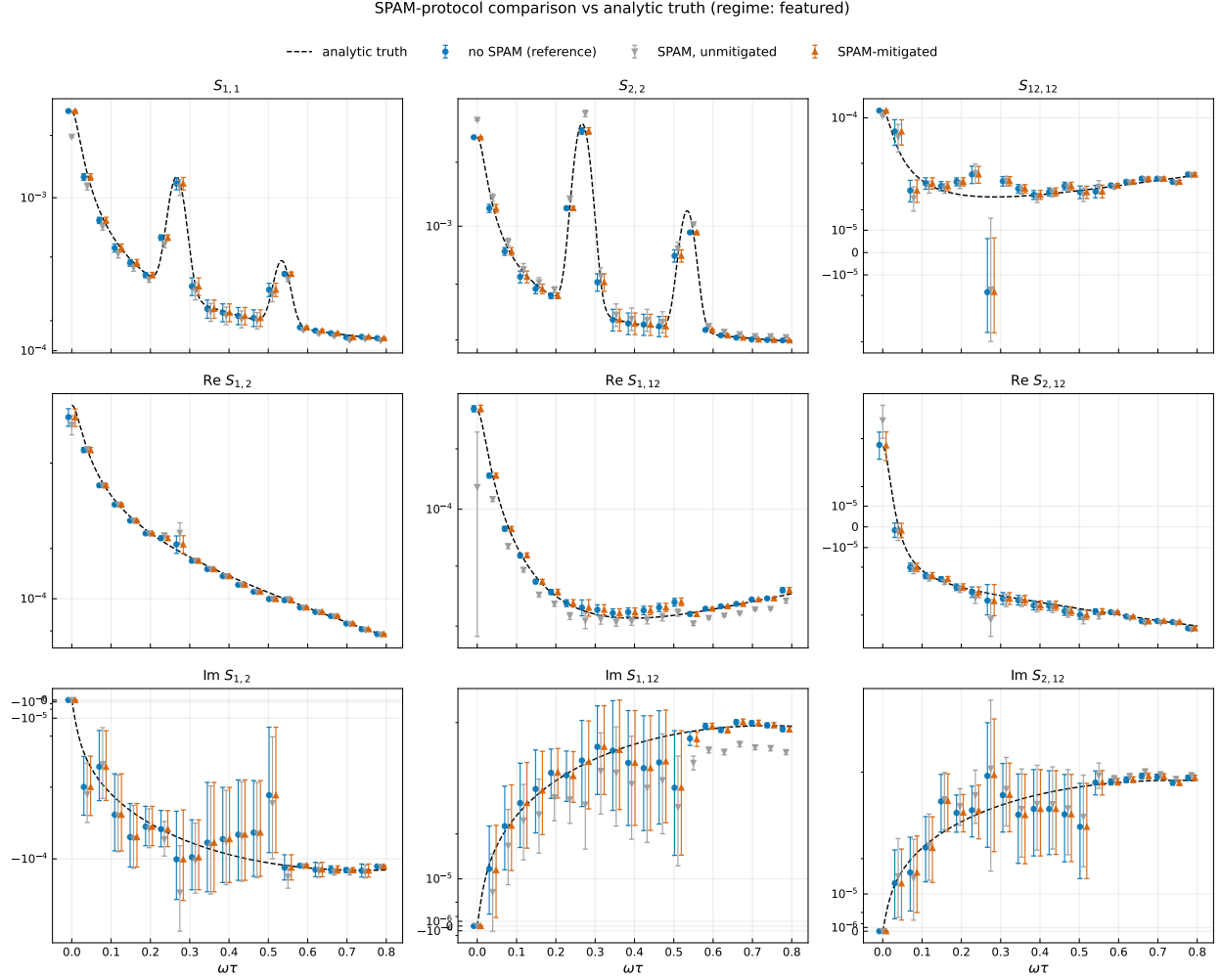


Figure 2: SPAM-protocol comparison vs. analytic truth (64k shots, featured regime): reference (no SPAM), raw (SPAM, unmitigated), and SPAM-mitigated arms.

knowledge	$T_G = 80\tau$		$T_G = 320\tau$	
	NT true $1 - F_{\text{pro}}$	vs. all-6	NT true $1 - F_{\text{pro}}$	vs. all-6
all 6 spectra	6.99e-3	1.00	4.29e-2	1.00
robust set (4): no $S_{1,12}, S_{2,12}$	6.99e-3	1.00	4.29e-2	1.00
diagonals (3): no crosses	6.99e-3	1.00	4.29e-2	1.00
1Q only (2): $S_{1,1}, S_{2,2}$	7.44e-3	1.06	4.27e-2	1.00
best CDD (for scale)	1.17e-2		7.42e-2	

Table 3: Ask 2(b): true infidelity of the blind NT winner vs. the spectral knowledge given to the optimizer (identical restart initializations across rows).

4 Ask 3: why characterize SPAM?

Two regimes, now both measured:

Spectroscopy: SPAM bias is large and certain. At 64k shots the raw arm’s reconstruction is wrong at $14\text{--}34\sigma$ (Fig. 2) — any spectroscopic claim made from unmitigated data is falsified at high confidence. SPAM mitigation restores truth ($\equiv$ reference within bars), and the double-echo experiment turns $S_{12,12}(0)$ from a bounded into a determined quantity (1.12×10^{-4} vs. truth 1.11×10^{-4}).

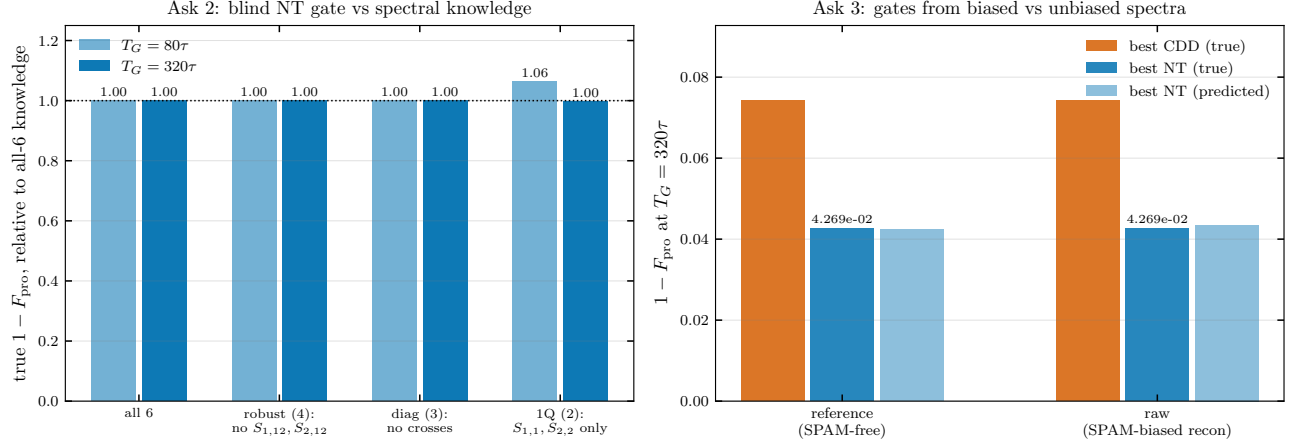


Figure 3: Left (ask 2): the blind NT gate is insensitive to removing cross- and even ZZ-spectral knowledge from the optimizer. Right (ask 3): a gate optimized on the raw (SPAM-biased) reconstruction performs identically to one optimized on the SPAM-free reference; the cost of SPAM lives in the spectroscopy and the prediction, not the design.

Gates: the design is forgiving; the prediction is not. Optimizing on the raw arm’s biased spectra produces a gate *identical in true performance* to the reference-arm gate at $T_G = 320\tau$ (Fig. 3, right): both arms find the same NT(148,150) sequence at true $1 - F_{\text{pro}} = 4.27 \times 10^{-2}$. The smooth SPAM bias does not move the optimal filter placement. What SPAM characterization buys at the gate level is the *credibility of the predicted numbers*: the reference arm predicts its gate to 0.7%, the raw arm to 1.9% at this gate time — and the uncertainty bands of §5 are only meaningful with unbiased spectra and honest error bars.

Synthesis with ask 2: since the robust protocol’s lost cross-spectra do not affect the gate design, **SPAM-robust QNS + NT optimization is a self-sufficient pipeline** when one accepts budget-level (not channel-resolved) error accounting; SPAM-mitigated QNS recovers the full six-spectrum budget.

5 Gates on the anchored model: the honest headline

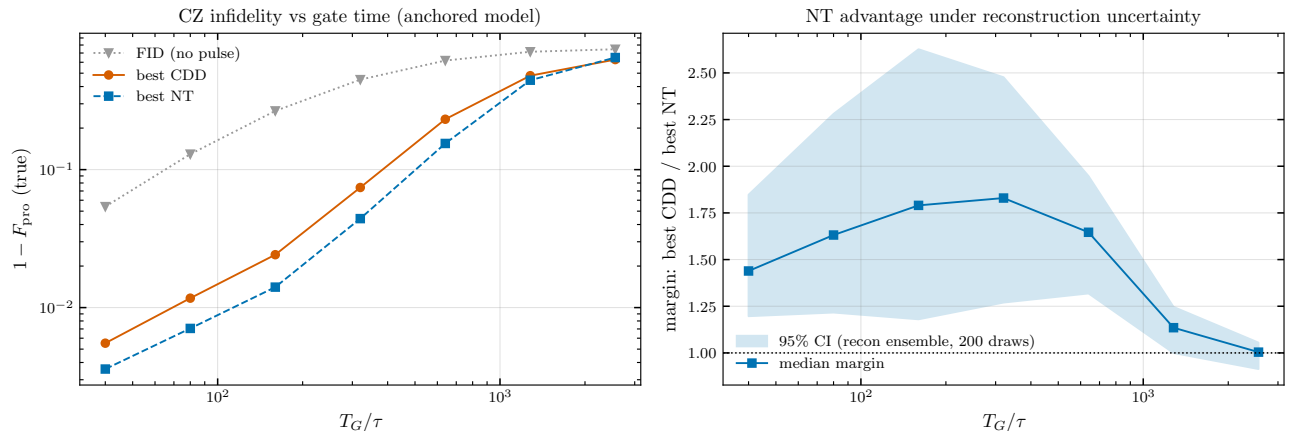


Figure 4: Left: CZ process infidelity vs. gate time on the anchored featured model — FID, best CDD (blind-selected), best NT (blind-optimized), all scored on truth. Right: the NT-vs-CDD margin with its 95% CI under the reconstruction uncertainty ensemble (200 draws, stat + systematic bars, coherent-systematic model, tail-fit sensitivity included).

operation	published (main_v9, superseded model)			regenerated (anchored model)		
	FID	best CDD	best NT	FID	best CDD	best NT
idling 640τ	5.13e-2	3.67e-5	7.06e-5	6.17e-1	7.29e-2	5.34e-2
idling 2560τ	1.91e-1	2.55e-3	5.81e-4	7.46e-1	3.54e-1	2.81e-1
entangling 320τ	4.75e-1	1.43e-1	4.40e-4	4.49e-1	7.42e-2	4.41e-2

Table 4: The published sweet-spot numbers vs. the anchored-model regeneration. The published 10^2 – 10^3 NT suppressions were artifacts of the superseded synthetic spectra (negligible weight in the NT passband). Winner flips to NT in the short-idling row.

The corrected headline: **NT beats the best CDD at every operational gate time** — 1.5 – $1.8\times$ for the entangling gate across $T_G = 40$ – 640τ and 1.3 – $1.4\times$ for idling — **with 95% CI lower edges $\geq 1.18\times$ under the full reconstruction-uncertainty ensemble**, degrading gracefully to parity only beyond T_2 . The structural argument stands independently: decoupling promotes the two-qubit channels to 20% of the CDD error budget, and NT sequences shape y_{12} to address exactly that.

NT-vs-CDD margins (true, with predicted 95% CI): 40τ : $1.54\times [1.20, 1.85]$; 80τ : $1.66\times [1.22, 2.28]$; 160τ : $1.72\times [1.18, 2.63]$; 320τ : $1.68\times [1.27, 2.48]$; 640τ : $1.50\times [1.32, 1.95]$; 1280τ : $1.08\times [1.00, 1.25]$; 2560τ : $0.97\times [0.91, 1.06]$.

6 The three answers, in talk form

Ask 1 — how much dephasing? $T_2^* = 20\mu\text{s}$ (800τ) anchored to purified ^{28}Si ; Gaussian FID $T_2 \approx 19\mu\text{s}$; gates studied span 0.05 – $3.3T_2$. Spectra: $1/f$ -type charge noise (exponents 0.4 – 0.7 , Si and GaAs agree), GaAs nuclear-difference line triplet at $1.7/1.7/3.4$ MHz, exchange noise at 10% amplitude coupling.

Ask 2 — do I need all six spectra? For the error budget, yes: decoupling promotes the two-qubit channels from 1% to 20% of the gate error. For the optimizer, no: the blind NT design is unchanged down to single-qubit knowledge ($\leq 1\%$ effect) — so the robust protocol’s four-spectrum set is design-complete, and two-qubit QNS is what makes the *prediction* and the budget honest.

Ask 3 — why SPAM? Unmitigated SPAM falsifies the spectroscopy at 14 – 34σ ; mitigation restores truth exactly (and the double-echo determines $S_{12,12}(0)$). The gate *design* survives SPAM bias — the penalty lives in prediction credibility, which is precisely what the paper quotes.

Provenance: QNS-2Q commits through 29dac8a (data) / c14358f (pipeline); noise model anchored-tarucha-20260610; runs: DraftRun_NoSPAM_featured (4k, recorded), DraftRun_SPAM_featured_{reference,raw,mitigated} (64k); all optimization blind, benchmarks exact (always-folded), reproducible seeds.